Calculus I Assignment 3

1. Evaluate the following limits.

(a) $\lim_{x\to 3}(x-1)^2(x+1)$

(b) $\lim_{x\to 5}\dfrac{x+3}{5-x}$

(c) $\lim_{x\to 3}\dfrac{9-x^2}{x-3}$

(d) $\lim_{x\to 5}\dfrac{x^2-3x-10}{x-5}$

(e) $\lim_{x\to -2}\dfrac{x^2-x-6}{x^2+3x+2}$

(f) $\lim_{x\to 4}\dfrac{\sqrt{x}-2}{x-4}$

(g) $\lim_{x\to\infty}\left(\sqrt{x^2+3}-x\right)$

(h) $\lim_{x\to -\infty}\left(1+\dfrac{3}{x}\right)^{2x}$

2. Find $\lim_{x\to\infty} f(x)$ and $\lim_{x\to -\infty} f(x)$. If the limiting value is infinite, indicate whether it is $+\infty$ or $-\infty$.

(a) $f(x)=x^3-4x^2-4$

(b) $f(x)=\dfrac{x^2-2x+3}{2x^2+5x+1}$

(c) $f(x)=\dfrac{3x^2-6x+2}{2x-9}$

3. Find the indicated one-sided limit.

(a) $\lim_{x\to 3+}\dfrac{\sqrt{x+1}-2}{x-3}$

(b) $\lim_{x\to 3-} f(x)$ and $\lim_{x\to 3+} f(x)$, where $f(x)=\begin{cases}2x^2-x & \text{if } x<3\\ 3-x & \text{if } x\geq 3\end{cases}$

4. The limit of a function can be completely described in terms of the limits of sequences. To do this, show that these two statements are true:

(a) If $\lim_{x\to c} f(x)=L$ and x_n is any sequence with $x_n\neq c$ and $\lim_{n\to\infty} x_n=c$, then $\lim_{n\to\infty} f(x_n)=L$.

(b) If $\lim_{n\to\infty} f(x_n)=L$ for every sequence x_n with $x_n\neq c$ and $\lim_{n\to\infty} x_n=c$, then $\lim_{x\to c} f(x)=L$.

Conclude that in the discussion of continuity, $\lim_{x\to c} f(x) = f(c)$ is equivalent to $\lim_{n\to\infty} f(x_n) = f(c)$ for every sequence x_n tending to c.

5. Using the definition of a limit, prove the following statements.

(a) $\lim_{x\to a} c = c$

(b) $\lim_{x\to 2} x^3 = 8$

(c) $\lim_{x\to 2} \sqrt{x} = \sqrt{2}$

(d) $\lim_{x\to -3} \dfrac{1}{(x+3)^4} = \infty$

6. Prove that $\lim_{x\to 0} \sin(x) = 0$ and $\lim_{x\to 0} \cos(x) = 1$.

7. Suppose that $f(x)$ is continuous at $x = 0$. Prove that $\lim_{x\to 0} xf(x) = 0$.

8. If the function f is defined by $f(x) = \begin{cases} 0 & \text{if x is rational} \\ 1 & \text{if x is irrational} \end{cases}$ prove that $\lim_{x\to 0} f(x)$ does not exist.

9. Show that the equation $\sqrt[3]{x-8} + 9x^{2/3} = 29$ has at least one solution for the interval $0 \le x \le 8$.

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